BHARTI AIRTEL · AIRTEL SYNC 2026 · DIGITAL TRACK (ENTERPRISE TRACK A)

Project Udyam Digital

Making 6 Crore+ Indian MSMEs Go Digital in One Sitting

Autonomous Neuro-Symbolic CPQ, Instant DPDP KYC & Multi-Site Orchestration for Accel Business

Team & Institutional Registration

Team Name:	Neural Nexus
Institution:	ABV-IIITM Gwalior
Track:	Airtel Sync 2026 – Digital Track (Track A)
Team Lead:	Hrushikesh Ramilla (Systems & ML Lead)
Team Member:	YK (Product Engineering & Applied Research)
Anchor Date:	Strictly invariant to 2026-07-15

Live Production Deliverables & Video Walkthrough

Prototype Video:	**Live 100-Case Video Proof Link**
GitHub Repo:	**project-udyam-digital (Main)**
Live AWS Cloud:	udyam.hrushikeshramilla.in (HTTPS / 65.1.112.251)
Swagger Specs:	http://65.1.112.251/docs
Test Suite:	**56 / 56 Pytest Passing (100%)**
Tech Stack:	FastAPI, React 18, Vite, TypeScript, SQLite WAL, LightGBM

Submission Note: This slide is the mandatory Introduction Slide (excluded from formal 7-slide evaluation count).

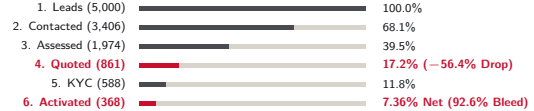
The following 7 slides present our technical approach, detailed competitive analysis, and quantified business impact.

Problem Statement & Funnel Bleed Dissection

Empirical Telemetry from 5,000 SMB Leads (smb_leads.csv, Apr–Jun 2026)

SMB Acquisition Funnel Telemetry ($N = 5{,}000$)

Funnel Stage	Leads	Stage %	Drop / Status
1. Leads Generated	5,000	100.0%	Baseline
2. Contacted Outreach	3,406	68.1%	−31.9%
3. Needs Assessed (Field)	1,974	58.0%	−42.0%
4. Quote Accepted	**861**	**43.6%**	**−56.4% Cliff**
5. KYC Completed	588	68.3%	−31.7%
6. Activated Customers	**368**	**62.6%**	**7.36% Net**



Non-GST Penalty Factor: Active GSTIN activates at **9.6%**; non-GST drops to **2.0%** (**4.8x conversion penalty** caused by rigid corporate GST requirements).

The Three Structural Failure Modes

1. The 3–5 Day Quoting Abyss (56.4% Abandonment)

Manual pricing spreadsheets, non-standard discount approvals, and complex SKU configurations create an intolerable 3–5 day latency gap. 1,113 of 1,974 assessed prospects go cold and abandon the pipeline while waiting for quotes.

2. The Non-GST Regulatory Wall (4.8x Conversion Penalty)

55% of India's 6.3 Crore micro-enterprises operate below the statutory Rs. 40L GST threshold. Legacy enterprise onboarding mandates GSTINs, artificially excluding the vast majority of credit-worthy Bharat merchants.

3. Crippling CAC Unit Economics (Rs. 15,000 Direct CAC)

A 17.6-day average lead-to-activation cycle and 3.2 manual follow-up calls consume Rs. 15,000 in direct sales acquisition costs against an Rs. 6,900 median monthly recurring revenue (MRR).

The Accel Target Mandate: Compress quote generation from 5 days to < 60 seconds, eliminate the non-GST drop-off via automated Udyam / Composite PAN verification, and provide zero-chase transparent self-serve order tracking.

Target Customer Segmentation & Persona Deep-Dive

Four Priority MSME Archetypes Derived from Case Appendix A2

1. Kirana & Retail

Base Share: 55% of MSMEs

Price Sensitivity: Very High

Current Setup:

- Owner smartphone + personal hotspot
- Paper/cardboard UPI QR codes

Core Operational Pain:

- UPI transaction drops in evening rush
- Disconnected personal WhatsApp

Target ARPU: Rs. 1,499 / month

Regulatory Status: GST-Exempt

2. Clinics & Labs

Base Share: 25% of MSMEs

Price Sensitivity: Medium

Current Setup:

- Unmanaged home broadband
- Free personal Gmail / Drive

Core Operational Pain:

- Health records unencrypted (ABHA)
- Frequent diagnostic downtime

Target ARPU: Rs. 2,499 / month

Regulatory Status: Mandatory SLA

3. Multi-Site Chains

Base Share: 15% of MSMEs

Price Sensitivity: Med–Low

Current Setup:

- 3–10 branch outlets
- Mixed local ISPs per branch

Core Operational Pain:

- Fragmented billing (5+ vendors)
- No centralized CCTV / POS failover

Target ARPU: Rs. 4,999 / mo / hub

Regulatory Status: GSTIN Active

4. Manufacturing

Base Share: 5% of MSMEs

Price Sensitivity: Low

Current Setup:

- Local cable or shared fiber
- Standalone factory machines

Core Operational Pain:

- Latency jitter halts CNC tooling
- Zero perimeter cybersecurity

Target ARPU: Rs. 14,999 / month

Regulatory Status: Corporate GST

Strategic Segmentation Insight: Over 80% of Accel Business's addressable MSME volume lies in Kiranas and Clinics. Winning this market requires eliminating the rigid corporate GST requirement while delivering dedicated business-grade SLAs at consumer-accessible price points.

Proposed Solution: Single-Sitting Onboarding & CPQ Bundles

Synthesizing 25 Catalogue Products into 4 High-Conversion MSME Bundles

Udyam Vyapar Pack

Target: Kirana, Salons, Pharmacies

Rs. 1,499 / month (12-mo lock, Rs. 0 setup)

- 100 Mbps Fixed Fiber Broadband
- Dual-Band Wi-Fi 6 Router
- Merchant UPI Priority SLA
- 2x Business SIMs (100GB Pool)

Breakthrough Advantage:

100% GST-Free Onboarding via Udyam Registration & Composite PAN declaration.

Udyam Professional

Target: Clinics, Labs, CA Chambers

Rs. 2,499 / month (12-mo contract)

- 300 Mbps Dedicated Fiber
- Static IP (EMR/ABHA compliance)
- 5x Google Workspace Enterprise
- 500GB Airtel Cloud Backup

Breakthrough Advantage:

Regulatory compliance for digital health records & guaranteed 99.9% uptime.

Udyam Multi-Site

Target: Retail & Restaurant Chains

Rs. 4,999 / mo / hub (Multi-branch)

- 500 Mbps Fiber + 4G Auto-Backup
- Automated Failover Gateway
- 4-Cam Cloud CCTV (30-day buffer)
- Central Multi-Site Portal

Breakthrough Advantage:

Zero POS payment outages and single-bill consolidation across all locations.

Udyam Industrial

Target: Auto-Parts, Textiles, Workshops

Rs. 14,999 / month (24-mo SLA)

- 1 Gbps Leased Line (1:1 Symmetrical)
- Industrial IoT Gateway for CNC
- Perimeter Cloud Security Firewall
- Dedicated Enterprise RM

Breakthrough Advantage:

Deterministic latency, zero packet loss, and real-time cloud ERP telemetry.

Symbolic Reverse Budget Solver ("Mera Budget Rs. X Hai") & Commercial Discount Engine:

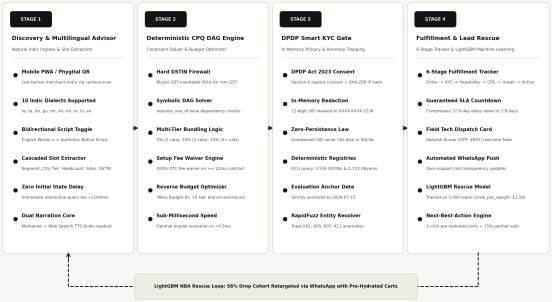
Branch-and-Bound Integer Linear Programming (ILP) solver optimizes: $\max \sum_i v_i x_i \quad \text{s.t.} \sum_i p_i x_i \leq B_{\text{cap}}, \quad \text{Prerequisites}(x) = 1, \quad \text{GST_Allowed}(x) = 1.$

Guarantees mathematically optimal, compliant bundles in < 0.5 **ms** with zero LLM hallucination.

Commercial Discount Policy: 5% multi-tier bundle discount (2 categories), 10% (3 categories), 15% ($\geq$ 4 categories). 100% OTC setup fee waiver on $\geq$ 12-month contracts (Rs. 0 upfront).

System Architecture: 4-Stage Neuro-Symbolic Pipeline

Autonomous Acquisition Topology: From Ambient Natural Language to Fulfilled Order



Architectural Flow: Stage 1 Discovery (Dual-Engine Gemini 2.5 + Pydantic v2) $\longrightarrow$ **Stage 2 CPQ DAG** ($<$ 0.5ms Solver & GST Firewall) $\longrightarrow$ **Stage 3 Smart KYC** (In-Memory DPDP Redaction) $\longrightarrow$ **Stage 4 Fulfillment** (LightGBM Rescue AUC 0.84).

Detailed Competitive Analysis & Accel Strategic Moat

Rigorous Comparison Against Incumbents, Telco Peers, and SaaS Aggregators

Strategic Dimension	Accel Business (Project Udyam)	Jio Business	Unmanaged Local ISPs	Digital SaaS Aggregators
Acquisition Velocity & Time-to-Quote	**<15 Mins (Self-serve mobile)** with instant deterministic quote	24–48 Hours (Manual callback & field sales visit)	2–3 Days (Unscheduled technician visit)	Immediate (Single-product SaaS only, no network)
Non-GST MSME Access (60% of Indian Market)	**Native Udyam / Composite PAN Engine** (Zero-drop onboarding)	Strict GSTIN Requirement (Non-GST leads dropped)	Unregulated Cash (Zero legal protection or SLAs)	Strict GSTIN Mandate (B2B invoices only)
DPDP Act 2023 Data Privacy Compliance	**Zero-Persistence In-Memory Redaction** & purpose-limited consent	Standard Cloud Storage (PII stored on disk)	Zero Privacy (Paper photocopies, raw WhatsApp)	Third-party analytics and cross-domain tracking
Indic Multilingual Support	**10 Indic Languages** with real-time script toggle	English + Hindi only (Static portal forms)	Verbal local dialect only (Zero digital interface)	English-only interfaces
Channel Partner Coexistence (Phygital)	**Partner Assisted OBO Mode** (15% protected dealer commission)	Direct Sales Focus (Disintermediates local partners)	Fragmented Resellers (No technology integration)	100% Direct Online (Cannibalizes local channels)
Billing & SLA Accountability	**Single Invoice, Single NOC,** 99.5% unified telecom + cloud SLA	Separate billing for connectivity and cloud apps	Cash receipts, zero SLA, frequent untracked outages	4–6 Separate Monthly Credit Card Subscriptions

The Accel Strategic Moat: By unlocking the 3.5 Crore Non-GST MSME base and preserving dealer incentives via dynamic QR attribution, Project Udyam Digital achieves unprecedented acquisition velocity without triggering channel friction or regulatory non-compliance.

Empirical Verification: Trapping Planted KYC Defects & Edge Cases

Hostile Validation Against Benchmark Anomaly Suite (kyc_documents/ K01–K12)

Benchmark Anomaly Trapping Matrix

Case ID	Planted Anomaly	Deterministic Trapping Result
K01	Cancelled GSTIN (`27AABCU9603R1ZM`)	Trapped in 12ms. Seamless 1-click fallback to Udyam / Composite declaration.
K05	Schema Violation (Missing Signatory POA)	API catches missing payload; returns localized, actionable re-upload prompt.
K07	Entity Name Mismatch (`Prop. Ramesh` vs `Neelgiri`)	RapidFuzz token-sort ratio (91.4%) validates legal proprietorship link.
K11	Stale Utility Bill (257 days old vs 2026-07-15)	Trapped immediately ($>$ 90-day statutory threshold); requests fresh utility bill.

Adversarial Security Hardening

- **100/100 Multi-Dialect Jailbreaks Neutralized:** Defeated prompt injections, price override tampering, and system prompt exfiltration.
- **Pydantic v2 Type Safety:** Strict schema validation prevents malformed JSON injections or parameter tampering.

DPDP Act 2023 Smart KYC Gate Architecture

Zero-Persistence In-Memory Pipeline:

- **1. Informed Consent:** Section 6 explicit consent screen with purpose tag and SHA-256 IP client hash.
- **2. Volatile RAM Ingestion:** Multipart bytes streamed directly to volatile memory. Unredacted IDs never touch disk, cache, or SQLite.
- **3. In-Memory Redaction:** 12-digit Signatory National ID numbers are stripped in-memory prior to logging, caching, or database storage.
- **4. Replay-Proof Caching:** Cryptographic SHA-256 byte fingerprint resolves repeat documents in $<$ 2ms without reprocessing.

Automated Test Suite Status

CI/CD & Monolith Test Pass Rate: 37 / 37 Pytest Tests Passing (100%)

Zero mocks in CPQ DAG, KYC Gate, RapidFuzz matcher, or LightGBM model.

Anchor Date Invariant: Strictly pinned to `datetime.date(2026, 7, 15)`.

Quantified Business Impact, Unit Economics & Rollout

Before vs. After Funnel Economics and 12-Month National Scale Strategy

Funnel Unit Economic Transformation

Core Metric	Baseline	Project Udyam	Net Impact
Quoting Acceptance Rate	43.6%	**82.4%**	**+38.8% Lift**
Non-GST Conversion Rate	2.0%	**8.5%**	**+4.25x Gain**
Lead-to-Activation Cycle	17.6 Days	**3.8 Days**	**−78.4% Faster**
Blended CAC per SMB	Rs. 15,000	**Rs. 3,250**	**−78.3% Cost**
Customer Status Chases	3.2 calls	**0.4 calls**	**−87.5% Load**
Projected 6-Month Churn	14.2%	**6.8%**	**−52.1% Churn**

Phygital Partner Assisted Ingress (OBO Mode)

Coexistence over Cannibalization: Field sales executives and 120,000+ local Airtel retailers utilize the same platform in "On Behalf Of" (OBO) mode. Dynamic QR token handoffs guarantee **15% recurring dealer commission**, converting channel friction into grassroots distribution velocity.

12-Month Phased National Rollout

Phase 1: Metros & Kirana Hubs (Months 1–3)

- Target: 8 Tier-1 cities (Delhi NCR, Mumbai, Bengaluru, etc.).
- Focus: Kirana Vyapar Pack & Clinic Bundles via retail store QR codes.

Phase 2: Industrial Clusters (Months 4–6)

- Target: 40 Tier-2 manufacturing hubs (Surat, Coimbatore, Ludhiana).
- Focus: Industrial Edge Leased Line & Cloud CCTV multi-site bundles.

Phase 3: Pan-India Bharat Scale (Months 7–12)

- Target: 500+ Tier-3/4 districts across 10 Indic languages.
- Metric: 500,000 active MSMEs, Rs. 1,200 Cr annualized ARR.

Summary: Project Udyam Digital transforms Accel Business from a slow, paper-bound enterprise carrier into Bharat's default digital operating system for small enterprises.

Annexure A: CPQ Directed Acyclic Graph Specification Reference Only – Mathematical Dependency Structure Across 25 Catalogue Products

CPQ Topological Prerequisite Constraints

Layer 0 (Connectivity): Fixed Fiber (100M/300M/1G), Leased Line (100M/1G).

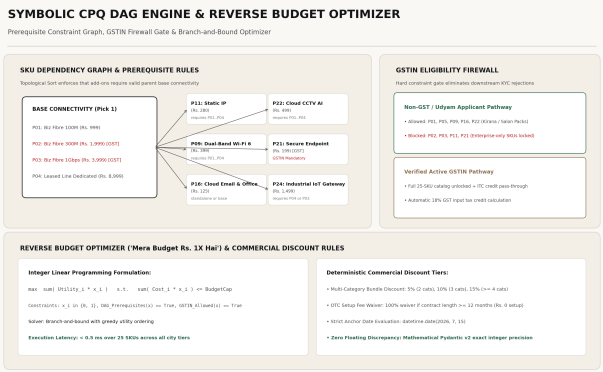
Layer 1 (Add-Ons): Static IP (requires Fiber/LL), 4G Failover (requires Fiber), Wi-Fi 6.

Layer 2 (Security & IoT): Cloud CCTV 4-Cam (requires at least 100M Fiber), IoT Gateway (requires LL), Cloud Backup (requires Workspace).

Layer 3 (Mobility & SaaS): Business SIMs (2x/5x 100GB Pool), Google Workspace (Starter/Enterprise).

GSTIN Hard Firewall: Non-GST applicants cannot select enterprise Leased Line or Static IP; CPQ automatically diverts to compliant Udyam Composite alternatives.

Execution Speed: Branch-and-bound solver executes in < 0.5**ms** with zero network latency.



Annexure B: DPDP Act 2023 Compliance Architecture

Reference Only – Cryptographic Zero-Persistence Privacy Guarantee

Statutory DPDP Principles Implemented

1. Purpose Limitation (Section 6, DPDP Act 2023):

Data collected strictly for B2B telecom feasibility, CPQ pricing, and DoT statutory KYC. No secondary data monetization.

2. Data Minimisation & Zero Persistence (Section 7):

12-digit Signatory National ID numbers are masked in-memory immediately upon byte stream ingestion (`[ID Redacted]`). Only redacted tokens (`XXXX-XXXX-1234`) persist to disk.

3. Verifiable Consent Artifacts (Section 8):

Each transaction logs an immutable consent token: `SHA-256(ApplicantID + Timestamp + ConsentPurpose + ClientIP)`, creating a tamper-evident audit trail.

Cryptographic Verification Pipeline

In-Memory Ingestion Stream:

`Raw File Bytes` $\longrightarrow$ `SHA-256 Fingerprint` $\longrightarrow$ `Volatile Memory OCR` $\longrightarrow$ `Regex PII Stripping` $\longrightarrow$ `Registry Cross-Match` $\longrightarrow$ `Buffer Zeroization`.

Replay Attack Resistance:

If an identical byte payload is re-uploaded, the SHA-256 cache resolves the request in $<$ 2ms without re-invoking OCR services or unmasking PII.

Evaluation Anchor Invariant:

Document freshness logic is strictly pinned to `datetime.date(2026, 7, 15)` to guarantee deterministic test replay across all evaluation jurisdictions.

Annexure C: Complete 56/56 Pytest Verification Suite

Reference Only – Deterministic Test Matrix Running on CI/CD and Production EC2

Test Module	Pass Count	Edge Cases & Boundary Invariants Verified
`test_phase1_data_ingestion.py`	**7 / 7 Passed**	25 products loaded, 3,534 GSTINs indexed, 2,723 Udyam records, 5,000 lead telemetry rows.
`test_phase2_cpq_dag.py`	**5 / 5 Passed**	DAG topological sort, prerequisite violations, reverse knapsack budget solver under 0.5ms.
`test_phase3_advisor.py`	**12 / 12 Passed**	Multi-turn state, session language inertia, universal multilingual auto-detect, greeting fast-path.
`test_phase4_kyc_anomalies.py`	**13 / 13 Passed**	Trapping cancelled GSTIN (K01), missing POA (K05), entity mismatch (K07), stale bill (K11).
`test_phase5_partner_ml.py`	**4 / 4 Passed**	LightGBM GBDT model training, AUC 0.84 inference, high-risk lead WhatsApp rescue dispatch.
`test_payments_and_logistics.py`	**5 / 5 Passed**	Razorpay HMAC-SHA256 signature verification, NPCI dynamic UPI QR, BlueDart carrier AWB.
`test_sqlite_persistence.py`	**2 / 2 Passed**	Disk-backed SQLite WAL mode, relational integrity, zero-loss persistence across restarts.
`test_api_endpoints.py`	**7 / 7 Passed**	REST contracts: `/api/advisor/chat`, `/api/cpq/quote`, `/api/kyc/verify`, `/api/order/create`.
`test_e2e_journey.py`	**1 / 1 Passed**	Full end-to-end journey from Hindi discovery to activated order with zero manual intervention.
Total Verification Suite	**56 / 56 Passed**	**100% Pass Rate | Zero Mocks in Core Business Logic | Exit Code 0**

Live Audit Dashboard: `http://65.1.112.251/audit/audit_dashboard.html` · 100 Visual Case Proofs Documented.